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Case report

Determination of glyphosate and AMPA in blood and urine from humans: About 13 cases of acute intoxication

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ABSTRACT

Acute intoxications after ingesting glyphosate are observed in suicidal or accidental cases. Despite low potential toxicity of this herbicide, a number of fatalities and severe outcomes are reported. Indeed, some authors have described the clinical features associated with blood and urine concentrations following intoxication. The purpose of this study is to describe the clinical feature and determinate the utility of the glyphosate concentration in blood and urine and the dose taken for predicting clinical outcomes. In 13 glyphosate poisoning cases treated in our laboratory within 7 years period from 2002 to 2009, we registered clinical observations and collected blood and urine samples to HPLC–MS–MS analysis. We classified our patients by the intoxication severity using simple clinical criteria. We obtained clinical observations from 10 patients and the others three patients were treated in forensic cases. Among the 10 patients, one was asymptomatic, 5 had mild to moderate poisoning and 2 had severe poisoning. There were 6 deaths whose 3 were forensic cases. The most common symptoms were oropharyngeal ulceration (5/10), nausea and vomiting (3/10). The main altered biological parameters were high lactate (3/10) and acidosis (7/10). We also noted respiratory distress (3/10), cardiac arrhythmia (4/10), hyperkaleamia, impaired renal function (2/10), hepatic toxicity (1/10) and altered consciousness (3/10). In fatalities, the common symptoms were cardiovascular shock, cardiorespiratory arrest, haemodynamic disturbance, intravascular disseminated coagulation and multiple organ failure. Blood glyphosate concentrations had a mean value of 61 mg/L (range 0.6–150 mg/L) and 4146 mg/L (range 690–7480 mg/L) respectively in mild–moderate intoxication and fatal cases. In the severe intoxication case for which blood has been sampled, the blood glyphosate concentration was found at 838 mg/L. Death was most of the time associated with larger taken dose (500 mL in one patient) and high blood glyphosate concentrations. To predict clinical outcomes and to guide treatment support in patients who ingested glyphosate, blood concentrations of this compound and the taken dose have been useful.

1. Introduction

Glyphosate is an herbicide widely used for agriculture. According to the classification of the World Health Organization, glyphosate is classified "U" (presents Unlikely to acute hazard in normal use) [1]. The incidence of intoxication to glyphosate in France appears to be low and true incidence is unknown. In our laboratory, glyphosate poisoning cases represent 12/40 of all cases for which pesticides analyses were required over a 7 years period from 2002 to 2009. Some cases of suicidal attempt by ingesting this substance have been reported and such ingesting can be lethal [2,3]. Clinical feature after ingesting glyphosate and its concentration in blood and urine has been described by several authors [4,5]. Glyphosate-poisoning is characterized by various symptoms such as gastrointestinal symptoms, altered consciousness, hypotension, respiratory distress, metabolic acidosis and renal failure [6–8]. Glyphosate formulation contains surfactants that probably enhance its toxicity. The mortality rate due to glyphosate poisoning is reported at 3.2% in a study included 601 patients with glyphosate acute poisonings [7]. In the same study, the authors reported that death was strongly associated with greater age, larger ingestions and high plasma glyphosate concentration on admission (>734 μg/mL).

Several methods have been described for determination of glyphosate in human plasma by using nuclear magnetic resonance (NMR) spectroscopy, gas chromatography coupled with mass spectrometry (GC–MS), capillary electrophoresis, liquid chromatography coupled with mass spectrometry (LC–MS), with UV detection (LC–UV) and with fluorescence detection (LC–FD) after derivatization and automated aminoacid analyser [11–15]. For example, Tomita and Okuyama described a sensitive and rapid

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procedure for determination of glyphosate and its metabolite aminomethyl phosphonic acid (AMPA) in serum after conversion into its p-toluenesulphonyl derivative and allowing the monitoring of glyphosate poisoning [16].

The purpose of this study is to describe the clinical feature in 13 cases treated in our laboratory, to compare those cases with the literature and to establish a likely relation between clinical severity and blood glyphosate concentration.

2. Materials and methods

2.1. Clinical cases

Case 1: A 69 year old man was admitted to the hospital for a suicide attempt. He had ingested a mouthful of herbicide formulation, Roundup® (glyphosate). His blood pressure, cardiopulmonary examination and neurological state showed no abnormalities. He presented a red throat and furred tongue without pharyngeal oedema. He had an erythema in the mucous membrane of arytenoids. The biological results were normal and the evolution was favourable. Glyphosate and AMPA were quantified in the serum taken 24 h after the ingestion.

Case 2: The patient, a 44 year old man, presented vomiting 30 min after having ingested an herbicide formulation containing glyphosate: Pistol EV®. His Glasgow coma scale (GCS) was 13 and he presented a mild bilateral mydriasis. Thereafter, he developed a dysrhythmia, altered consciousness and hypotonia that required intubation and ventilation. On his arrival at the hospital, the examination revealed a GCS at 3, severe mydriasis and a tachycardia. Arterial blood gas analysis displayed a severe metabolic acidosis (pH 6.65) and a marked hyperkalaemia (6.8 mmol/L) was noted. So a haemodialysis was carried out. After haemodialysis, his clinical state was not improved and it was noticied a refractory metabolic acidosis (pH 7.12), a severe anion gap (40 mmol/l) and a hypokalaemia (2.6 mmol/L). Soon after, he had a cardiovascular arrest. Resuscitation was attempted but the patient died. Serum samples were collected respectively 4.5 h and 6 h after the ingestion and 6 h after haemodialysis.

Cases 3, 4 and 5: No clinical observations were available on these cases which were in forensic contexts. We received whole blood and urine samples in case 3, whole blood, plasma and gastric contents in case 4 and whole blood, urine and serum in case 5.

Case 6: A 51 year old man was admitted to the hospital after a suicide attempt by ingesting 2 large glasses of herbicide, Roundup® (glyphosate). He was in shock (blood pressure less than 90 mmHg) and presented metabolic acidosis, respiratory distress and a disseminated intravascular coagulation. The patient's condition progressively deteriorated with a haemodynamic disturbance and a multiple organ failure. He died one day after hospitalization. Peripheral blood was collected 7 h after the ingestion for toxicological analysis.

Case 7: A 64 year old woman ingested a glass of Roundup® (glyphosate). On admission, she presented respiratory distress and hypersialorrhea requiring intubation. She developed cardiac shock, dysrhythmia caused by hyperkalaemia, metabolic acidosis, erosion of the digestive tract and hepatic toxicity. Clinical improvement was observed slowly and she was discharged 12 days after the hospitalization.

Case 8: After voluntary ingesting 100 mL of an unknown herbicide, a 65 year old woman complained of vomiting, dysphagia, pharyngeal obstruction and hypersecretion (1 h after ingestion) without consciousness disorder. Four hours later, she suffered from dysphonia. She was in shock (blood pressure 80 mmHg) and was intubed and ventilated. She developed profuse diarrhoea 8 h after ingestion. The biological investigation revealed a metabolic acidosis. Electrocardiogram displayed a ventricular block. Recuperation was slow but the patient was discharged 9 days after admission. Urine sample was collected less than 24 h after the ingestion.

Case 9: A 46 year old woman ingested a glass of Glyper® (glyphosate) in a suicidal attempt. Initial complaints were vomiting and diarrhoea (20 min after taking the dose). She developed a sore throat, dysphonia, metabolic acidosis, sinusal tachycardia and dispersed pain. The vomiting contained blood probably caused by digestive bleeding. A plasma sample was taken for toxicological analysis.

Case 10: The patient, a 67 year woman, ingested unknown amount of Roundup® (glyphosate). Clinical state was stable and the biological examination was unremarkable. Serum sample was obtained 5.5 h after the ingestion.

Case 11: A 60 year old man tried to poison himself by ingesting Roundup® (glyphosate) or Grivolax® (paraquat). On arrival to the hospital, he was in shock (blood pressure 80/20 mmHg). Resuscitation was attempted. He presented sweating, pulmonary obstruction, an impaired renal function, metabolic acidosis and haemodynamic disturbance. The shock state was refractory to treatment and the patient died one day after admission. We collected blood serum, urine and gastric contents 8 h after ingestion.

Case 12: A 59 year old man voluntarily drank 2 formulations: Verdys® (glyphosate) and Decis® (deltamethrin). He was admitted to hospital. He presented a Glasgow coma scale of 15, a metabolic acidosis with high lactate (4.5 mmol/L). Twelve hours following admission, no abnormality was found in his clinical course. He was transferred in gastroenterology department. The collection of serum and urine samples was undertaken 24 h after the ingestion.

Case 13: A 65 year old man accidentally ingested a mouthful of herbicide (glyphosate). On arrival at the hospital, he presented sore throat and dysphagia without ulceration. Clinical and biological investigation revealed no other abnormality. He was kept under observation and he was discharged on the second day of hospitalization.

Pistol EV® herbicide ingested by the patient 2 contained glyphosate (250 g/L) and diflufenicanil (40 g/l). The toxicity is almost related to glyphosate because diflufenicanil is not toxic: DL50 in rat $>$ 2500 mg/kg. In the other intoxication cases, we observed also ingestion of glyphosate with the following commercial formulations: Roundup® (5 cases), Verdys® (one case) and Glyper® (one case). All these last herbicide formulations contained only glyphosate in concentration of 360 g/L.

In case 10, we detected in addition of glyphosate a trace of deltamethrin in plasma.

2.2. Analysis procedure

The analysis procedure was inspired by the one described by Tomita and Okuyama with slight modifications [16]. Our method is based on a derivatization with Para-Toluene Sulfonyl Chloride (PTSCl) followed by a liquid–liquid extraction in acidic conditions in order to concentrate the derivative products. Dibutylphosphate is used as internal standard. Prior to the serum analysis, 2 mL of sample were deproteinized by adding 0.5 mL of acetonitrile. Briefly, derivatization was performed in 2 mL of urine or in 2 mL deproteinized serum by adding 0.5 mL of PTSCl (10 g/L in acetonitrile) and 1 mL of phosphate buffer at pH 11 for 15 min in an ultrasonic water bath then for 15 min at 50 °C.

PTSCl derivatives of glyphosate and AMPA were extracted with 8 mL of ethyl acetate after adding 1 mL of 6 M hydrochloric acid. Two μL of the dry residue reconstituted with 80 μL of 20 mM formate buffer (pH 3.0) and methanol (90/10, v/v) were injected into a high performance liquid chromatography coupled with tandem mass spectrometry system. The chromatographic system

Table 1
Blood and urine dosage in glyphosate acute poisoning.

		Blood concentration			Urine concentration			Sample time (h)	Age (years)	Taken dose (mL)
		Gly (mg/L)	AMPA (mg/L)	Ratio Gly/AMPA	Gly (mg/L)	AMPA (mg/L)	Ratio Gly/AMPA			
Mild and moderate symptomatology	Case 1	3.7	0.31	12	–	–		≈ 24 h	69	50
	Case 9	150	0.76	197	–	–		–	46	150
	Case 10	52.5	0.27	194	–	–		5.5 h	67	–
	Case 13	98.5	0.17	579	3000	9.3	322	1.5 h	63	50
	Case 12	0.678	ND	–	228	0.54	422	< 24 h	58	–
	Hori [5]	22.6	0.18	126	–	–		8 h	58	100
	Cartigny [4] *Case 3*	67.6	–		5750	–		< 24 h	–	–
Severe symptomatology	Case 7	838	1.3	645	13800	12.4	1112	25,5 h	64	150
	Case 8	–	–		17300	2.2	7863	< 24 h	65	100
	Cartigny [4] *Case 1*	2030	–		21100	–		–	40	20
	Cartigny [4] *Case 2*	1300	–		6100	–		< 24 h	46	–
	Motojyuku [10]	6645	15.1	440	–	–		–	56	400
	Cartigny [9]	8560	–		–	–		≈ 24 h	25	500
Fatalities	Case 3[a]	1550	24.6	63	22300	91.5	243	–	–	–
	Case 4[a]	2960	15.9	186	–	–		–	–	–
	Case 5[a]	690	2.94	235	998	2.49	400	–	–	–
	Case 6	7480	2	3740	–	–		7 h	51	500
	Case 2	5560	0.802	6933	–	–		4.5 h	44	–
	Case11	6640	15.4	431	14900	21	709	8 h	60	–
	Cartigny [4] *case 4*	321	–		–	–		–	–	–

Dose ingested is estimated as 50 mL: mouthful, 150 mL: glass, 250 mL: large glass.; ND: not detected.
[a] Case treated in forensic context, post-mortem samples.

consisted of a Perkin-Elmer Series 200LC pumping system, and a Series 200 auto-sampler. An API 2000 triple-quadripole mass spectrometer (AB Sciex, Courtaboeuf France) was used for detection with negative electrospray ionization mode. The following transitions (quantification transitions are underlined) were found to be optimal for the detection glyphosate $322.2 > 121.9/322.2 > 155.3/322.2 > 278.3$; and AMPA $264.0 > 63.0/264.0 > 81.1/264.0 > 79.0$. Validation procedure was performed according to both French Society of Analytical Toxicology recommendations [17] and proposed experiments, evaluations, and acceptance criteria for validation of new analytical methods to be used in single case studies of analysis of rare analytes [18]. The intra-assay precision and accuracy were assessed at low and high concentrations relative to calibration range: analysis of five spiked hair samples at each concentration. The detection limit (LOD) was defined as the lowest concentration giving a response of at least three-times the average of the baseline noise. The limit of quantification (LOQ) was defined as the lowest concentration that could be measured with an intra-assay precision CV% and relative bias less than 20%. This method exhibits following detection limits (LOD) and quantitation limits (LOQ), respectively, of 20 and 50 μg/L for glyphosate and AMPA. The two methods are linear until 2000 μg/L for both analytes. In addition, intra-assay precision CVs and relative bias are less than 20% over the calibrating range. Concerning matrix effects, no influence of interfering compounds on the signal were observed for all the analytes: no significant signal loss or increase was observed at the retention time windows of the compounds of interest and the I.S. Moreover and due to the very high concentration found in the real samples, high dilution of samples in deionised water was used. As a consequence potential matrix effects were suppressed. Chromatographic separation was performed at ambient temperature on a Nucleodur C18, 5 μm (150 mm × 1 mm I.D.) column (Macherey Nagel, Düren, Germany), using a linear gradient of acetonitrile (ACN) in 10 mmol/L, pH 3.0 ammonium formate buffer as mobile phase. Constant flow-rate 50 μL/min was programmed as follows: 0–1 min, 20% ACN; 1–9 min, 20–75% ACN; 9–9.5 min decrease from 75 to 20% ACN; 9.5–17 min, column equilibration with 20% ACN. All chromatographic solvents were degassed with helium beforehand.

3. Results and discussion

Table 1 presents patients glyphosate concentrations in blood and urine. These patients were divided in 3 groups by clinical features. We classified clinical features of glyphosate poisonings as mild, moderate and severe toxicity according to the clinical guide to severity of glyphosate outlined by Talbot et al., we joined mild and moderate in group 1 and we disjoined fatalities (group 3) to severe toxicity (group 2) [19].

Table 2 details clinical features and glyphosate concentration in 3 groups reported in literature and in our study [6,23,24]. Following an acute intoxication by glyphosate, the clinical signs most frequently described in our study (10 cases) were oropharyngeal ulceration (5/10) (sore throat, dysphagia), erosion of the gastro-intestinal mucous membrane (2/10) (gastric pains, blood vomiting) and vomiting (3/10). The principal disturbed biological parameters were: high lactate (3/10) and acidosis (7/10). In the serious forms were observed: respiratory distress (3/10), cardiac arrhythmia (4/10), hyperkaleamia, impaired renal function (2/10), hepatic toxicity (1/10), altered consciousness (3/10) and haemodynamic complication which in some cases necessitated a mechanical ventilation and intubation. In fatalities, cardiovascular shock (3/10), cardiorespiratory arrest, haemodynamic disturbance, intravascular disseminated coagulation (1/10) and multiple organ failure were observed.

As a result, we found a strong symptomatology-blood glyphosate concentration correlation (Fig. 1). In benign or moderate symptomatology, the blood concentrations of glyphosate observed and described in literature do not exceed 150 mg/L (Table 1). In severe symptomatology: the blood concentrations of glyphosate observed and described in litterature are usually higher than 1000 mg/L. In the 3 fatalities (treated in the service/non forensic cases), we noted blood concentrations of glyphosate higher than

Table 2
clinical features and biological parameters in intoxication by glyphosate.

	Our study				Other studies		
	Mild and moderate (n = 5)	Severe (n = 2)	Death (n = 6)	Total (n = 13)	Lee et al. [6]	Baselt et al. [24]	Hung et al. [22]
Age	60.6 ± 9.1	64.5 ± 0.7	51 ± 8 (n = 3)	58.7 ± 8.8	48 ± 2		56
Sexe ratio M/F	3/2	0/2	6/0	9/4	69/62	4/0	1/0
Taken dose[a] (mL)	83 ± 57 (n = 3)	125 ± 35 (n = 2)	500 (n = 1)	207 ± 188	138 ± 12		400
Nausea and or vomiting	1/5	1/2	1/3	3/10	93/126	1/2	
Sore throat	3/5	2/2	0/3	5/10	101/127	1/2	
Diarrhoea	1/5	1/2	0/3	2/10	26/131	–	
Altered consciousness	0/5	0/2	3/3	3/10	29/131	–	1
Respiratory distress	0/5	1/2	2/3	3/10	30/131	1/2	1
Ulceration of gastrointestinal tractus	1/5	1/2	0/3	2/10	–	–	
Arythmia	0/5	2/2	2/3	4/10	15/81	1/2	
Shock	0/5	0/2	3/3	3/10	13/130	–	1
Metabolic acidosis	2/5	2/2	3/3	7/10	29/81	1/2	
Low serum bicarbonate	0/5	0/2	2/3	2/10	39/81	–	
Hight serum Lactate	1/5	0/2	2/3	3/10	–	–	
Hepatic toxicity	0/5	1/2	0/3	1/10	–	–	
Renal failure	0/5	0/2	2/3	2/10	8/127	1/2	
Suicide attempt	4/5	2/2	6/6	12/13	116/131	3/4	1
Blood glyphosate (mg/L) [ranges]	61,07 ± 63,87 [0.6–150] (n = 5)	838 (n = 1)	4146 ± 2808 [690–7480] (n = 6)	2168 ± 2810 (0.6–7480)	–	929 ± 905 (67–2030) (n = 4)	6645
Blood AMPA (mg/L) [ranges]	0.302 ± 0.28 [0–0.79] (n = 5)	1.3 (n = 1)	10,27 ± 9.7 [0,8–24,6] (n = 6)	5.37 ± 8,33 [0–24.6]	–	–	15.1
Urinary glyphosate (mg/L) [ranges]	1614 ± 1960 [228–3000] (n = 2)	15550 ± 2474 [13800–17300] (n = 2)	12732 ± 10815 [2.49–91.5] (n = 3)	10360 ± 8828	–	10983 ± 8763 [5750–21100] (n = 3)	

We defined renal toxicity as elevation of creatinine > 120 μmol/L and hepatic toxicity as elevation of transaminase > 2 N; high lactate > 2.2 mmol/L, low bicarbonate < 24 mmol/L. Taken dose (a): by looking at the clinical files, we defined a mouthful as 50 mL, a glass as 150 mL and a large glass as 250 mL.

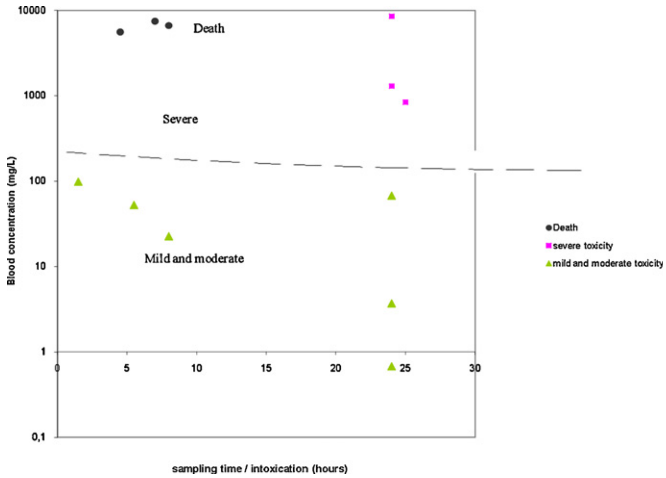

Fig. 1. Blood concentration of glyphosate in 12 cases reported in literature and in our study according to sampling time and gravity.

5000 mg/L. However, in forensic cases, the redistribution of glyphosate in *post-mortem* and sample late time can explain the different concentration described (690–2960 mg/L). As an illustration, chromatograms obtained from one fatal case are depicted in Fig. 2.

Fig. 1 is a representation of glyphosate blood concentrations observed in 12 poisoning cases reported in literature and in our study according to the time of ingestion and to the gravity [4,5,8,9]. We considered a timing of 2 h for sampling "on admission" and 36 h for "the 2nd day of the hospitalization".

Taken dose is in relation with the clinical course. In fact, in one fatal case, 500 mL had been ingested and in 5 survivors, the mean dose had been estimated at 100 mL (Table 1). Our findings are consistent with those of Lee et al. [6]. They estimated mean fatal ingested dose at 330 mL while survivors had a mean dose of 122 mL. The oral LD_{50} value of the Roundup® fluid was estimated to 2 mL/kg in humans by Suzuki and Watanabe [20].

Contrary to what was announced by Hori et al., who described ratios of concentrations glyphosate/AMPA of 126/1; 147/1 and 148/1 in 2 serums and 1 urine respectively, we did not manage to define a tendency of these average ratios of concentration, neither by matrix, nor by the degree of intoxication [5]. Indeed, the values which we obtained, in all the matrices and all the cases, range between 12/1 and 7864/1 (median 410/1, average 1349/1, standard deviation 2357/1). Even if the ratios described and observed were very different, we can conclude that glyphosate is very slightly metabolized in AMPA.

According to Testud and Grillet, in some pesticide formulations like Roundup®, the surfactant (polyoxyethylenediamine: POEA) would be at the origin of digestive disorders, esophagous caustic lesions, stomach and laryngeal lesions and aspiration pneumonitis [21]. In some reports, authors described that toxicity of glyphosate is enhanced by the surfactant contained in some formulations. For example, Roundup® fluid ingested in 5 cases contained 41% of glyphosate, 15% of anion surfactant (POEA), yellow-brown colour and odourless fluid at pH 4.8. Some authors suggested that gastrointestinal complications related to the corrosive effects of glyphosate [6]. Testud reports that hypotension can be explained by metabolic acidosis and hypovolemy, which has occurred following digestive losses (vomiting, diarrhoeas) [21]. Also hypoperfusion and the direct toxicity of the POEA can be at the origin of an impaired renal function and cardiac attack [21]. Hung et al. showed that laryngeal injury is the major cause of respiratory aspiration considered as a serious complication of glyphosate-surfactant intoxication [22].

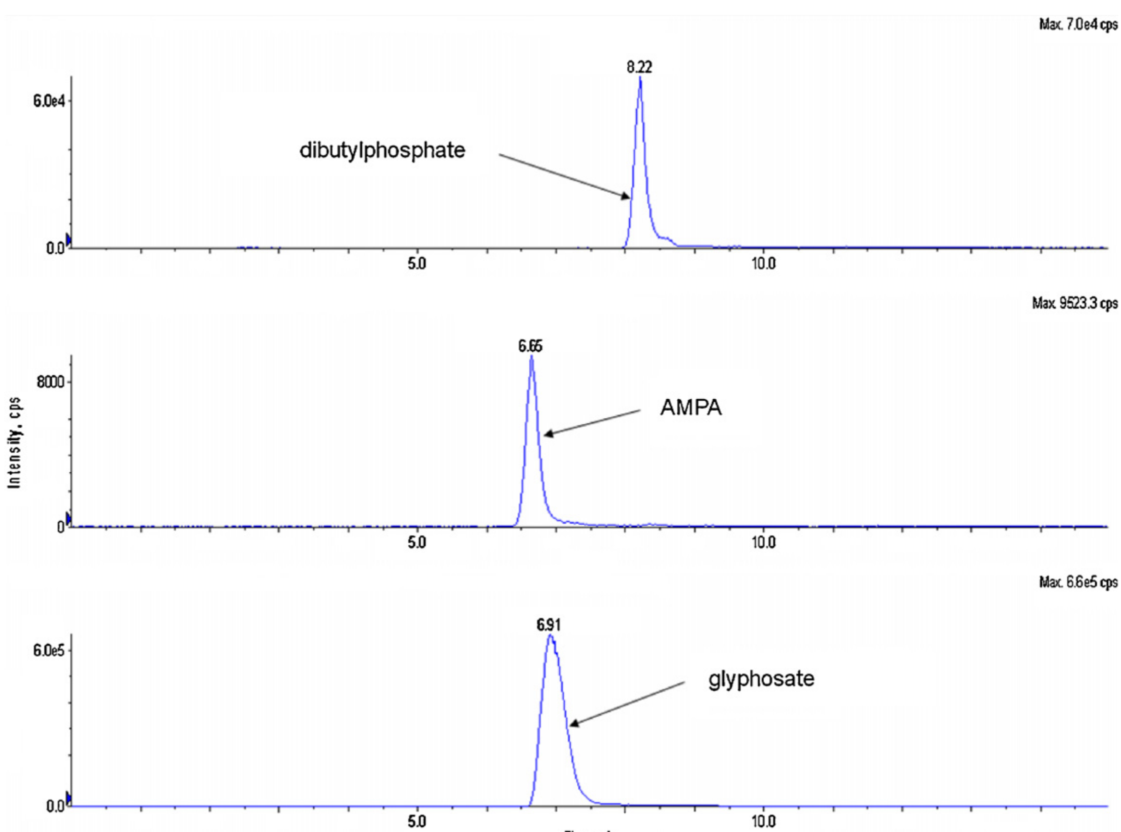

Fig. 2. Chromatogram of glyphosate 2960 mg/L and AMPA 15.9 mg/L in a diluted 1/100 plasma (case 4).

4. Conclusion

After intoxication by glyphosate ingestion, severe clinical signs (respiratory distress, cardiac arrhythmia) can be observed. A large amount of herbicide ingested (more than 190 mL) and glyphosate blood level higher than 734 mg/L can be considered predicting of gravity as described by Darren et al. [25].

Finally, we conclude that after a glyphosate poisoning especially in emergency clinical setting, the blood glyphosate concentration and the dose ingested correlates with the clinical feature and can be useful to evaluate the gravity.

References

[1] J.T. Stevens, D.D. Summer, Herbicides, in: W.J. Hayes, E.R. Laws (Eds.), Handbook of Pesticide Toxicology, Academic Press, San Diego, 1991, pp. 1339–1340..
[2] F.W. Sorensen, M. Gregersen, Rapid lethal intoxication caused by the herbicide glyphosate-trimesium (Touchdown), Hum. Exp. Toxicol. 18 (1999) 735–737.
[3] J. Stella, M. Ryan, Glyphosate herbicide formulation a potentially lethal ingestion, Emerg. Med. Australas. 16 (2004) 235–239.
[4] B. Cartigny, N. Azaroual, M. Imbenotte, Determination of glyphosate in biological fluids by ^{1}H and ^{31} P NMR spectroscopy, Forensic Sci. Int. 143 (2004) 141–145.
[5] Y. Hori, M. Fujisawa, K. Shimida, Determination of the herbicide glyphosate and its metabolite in biological specimens by gas chromatography-mass spectrometry a case of poisoning by roundup herbicide, J. Anal. Toxicol. 27 (2003) 162–166.
[6] H.L. Lee, K.W. Chen, C.H. Chi, Clinical presentations and prognostic factors of a glyphosate-surfactant herbicide intoxication: a review of 131 cases, Acad. Emerg. Med. 7 (2000) 906–910.
[7] D.M. Roberts, N.A. Buckley, F. Mohamed, A perspective observation study of the clinical toxicology of glyphosate-containing herbicides in adults with acute self-poisoning, Clin. Toxicol. 48 (2010) 129–136.
[8] R.L. Tominack, G.Y. Yang, Taiwan national poison center survey of glyphosate-surfactant herbicide ingestion, J. Toxicol. Clin. Toxicol. 29 (1991) 91–109.
[9] B. Cartigny, N. Azaroual, M. Imbenotte, Quantitative determination of glyphosate in human serum by ^{1}H NMR spectroscopy, Talanta 74 (2008) 1075–1078.
[10] M. Motojyuku, T. Saito, K. Akieda, Determination of glyphosate, glyphosate metabolites, and glufosinate in human serum by gas chromatography–mass spectrometry, J. Chromatogr. B 875 (2008) 509–514.
[11] M. Tomita, T. Okuyama, Determination of glyphosate and its metabolite (aminomethyl) phosphonic acid, in serum using capillary electrophoresis, J. Chromatogr. 571 (1991) 324–330.
[12] A. Ghanem, P. Bados, Glyphosate and AMPA analysis in sewage sludge by LC–ESI-MS/MS after FMOC derivatization on strong anion-Exchange resin as solid support, Anal. Chem. 79 (2007) 3794–3801.
[13] K. Qian, T. Tang, Residue determination of glyphosate in environmental water samples with high-performance liquid chromatography and UV detection after derivatization with 4-chloro-3,5-dinitrobenzotrifluoride, Anal. Chim. Acta 635 (2009) 222–226.
[14] E. Mallat, D. Bacelo, Analysis and degradation study of glyphosate and of aminomethylphosphonic acid in natural waters by means of polymeric and ion-exchange solid-phase extraction columns followed by ion chromatography–post-column derivatization with fluorescence detection, J. Chromatogr. A 823 (1998) 129–136.
[15] F. Parrot, R. Bedry, Glyphosate herbicide poisoning: use of a routine aminoacid analyser appears to be a rapid method for determining glyphosate and its metabolite in biological fluids, Clin. Toxicol. 33 (1995) 695–698.
[16] M. Tomita, T. Okuyama, High-performance liquid chromatographic determination of glyphosate and (aminomethyl) phosphonic acid in human serum after conversion into p-toluenesulphonyl derivatives, J. Chromatogr. 566 (1991) 239–243.
[17] A. Gruson, B. Dingeon, M.F. Dreyfuss, Y. Gaillard, A. Le Bouil, M. Lhermitte, S. Mazzega, M. Moulsma, G. Pépin, M. Perrin, J. Plesse, O. Roussel, M.T. Thevenot, E. Vinner, M. Deveaux, Help for validation of analytical procedures in toxicology and therapeutic drug monitoring, Ann. Toxicol Anal. 17 (2005) 1–20.
[18] F.T. Peters, O.H. Drummer, F. Musshoff, Validation of new methods, Forensic Sci Int. 165 (2007) 216-224, F. Musshoff and B. Madea. New trends in hair analysis and scientific demands on validation and technical notes, Forensic Sci Int. 165 (2007) 204–215.
[19] A.R. Talbot, M.H. Shiaw, J.S. Huang, Acute poisoning with a glyphosate-surfactant herbicide (Roundup): a review of 93 cases, Hum. Exp. Toxicol. 10 (1991) 1–8.
[20] O. Suzuki, K. Watanabe, Drug and Poisons in Humans A Handbook of Practical Analysis, Spinger-Verlag, Berlin, Heidelberg, 2005.
[21] F. Testud, J.P. Grillet, Produits phytosanitaires: intoxications aiguës et risques professionnels, Editions ESKA, Paris, 2007.
[22] D.Z. Hung, J.F. Deny, T.C. Wu, Laryngeal survey in glyphosate intoxication: a pathophisiological investigation, Hum. Exp. Toxicol. 16 (1997) 596–599.
[23] S.M. Bradberry, A.T. Proudfoot, J.A. Vale, Glyphosate poisoning, Toxicol. Rev. 23 (2004) 159–167.
[24] R.C. Baselt, Disposition of Toxic Drugs and Chemicals in Man, 7ème edition, Biomedical Publications, Foster City, CA, 2004.
[25] M.R. Darren, A.B. Nick, A prospective observational study of the clinical toxicology of glyphosate-containing herbicides in adults with acute self-poisoning, Clin. Toxicol. (Phila) 48 (2) (2010) 129–136.